



AUTHOR'S NAME

TITLE

Bachelor's/Master's Thesis

to achieve the university degree of

Bachelor/Master of Science

Bachelor's/Master's degree program: Author's program

submitted to

Graz University of Technology

Supervisor

Supervisor's name

Electric Drives and Power Electronic Systems Institute



Graz, Month XX, Year

Acknowledgements

This section is your opportunity to thank those who have helped and supported you personally and professionally during your thesis or dissertation process.

Abstract

An abstract is a short summary of a longer work. The abstract concisely reports the aims and outcomes of your research so that readers know exactly what your paper is about.

Kurzfassung

Eine Kurzfassung ist eine kurze Zusammenfassung deiner Arbeit. Die Kurzfassung berichtet prägnant über die Ziele und Ergebnisse deiner Forschung, sodass Leser genau wissen worum es in der Arbeit geht.

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Introduction

1.1 Inserting Figures, Subplots, etc.

To insert a figure in LaTeX, we need to include the `graphicx` package. This package is already included in our `packages.tex` file. In the following sections, you will learn how to insert figures in different ways.

1.1.1 Inserting a Single Figure

You can use the `figure` environment with the `\includegraphics` command to insert a single figure. Here's an example:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.5\textwidth]{path/to/your/image.png}
  \caption{Example figure}
  \label{fig:single}
\end{figure}
```



Figure 1.1: Example figure

Explanation of the individual commands:

- `\beginfigure`: Creates figure environment.
- `[htbp]`: Placement specifier (here, top, bottom, page).
- `\centering`: Centers the figure.

- `\includegraphics[width=0.5\textwidth]{path/to/your/image.png}`: Adds the image which is stored at *path/to/your/image.png* and adjusts the width of the image to 50% of the text width.
- `\caption{}`: Adds a caption below the figure.
- `\label{}`: Adds a label for referencing the figure.

This LaTeX-code will insert a single figure with the caption ‘Example figure’ and a label for cross-referencing. You can reference a figure using the `\ref{fig:label}` command, where you insert the actual label you gave the figure you want to reference. In Fig. 1.1, you can see how the code from above will look when applied in LaTeX.

1.1.2 Inserting Two Figures Side by Side

You can use the `subfigure` package to insert two figures side by side. Here’s an example:

```
\begin{figure}[htbp]
  \centering
  \begin{subfigure}[b]{0.4\textwidth}
    \includegraphics[width=\textwidth]{path/to/your/image-a.png}
    \caption{Figure A}
    \label{fig:subfiga}
  \end{subfigure}
  \hspace{0.05\textwidth}
  \begin{subfigure}[b]{0.4\textwidth}
    \includegraphics[width=\textwidth]{path/to/your/image-b.png}
    \caption{Figure B}
    \label{fig:subfigb}
  \end{subfigure}
  \caption{Two figures side by side}
  \label{fig:sidebyside}
\end{figure}
```

Explanation:

- `\begin{subfigure}[b]{0.4\textwidth}`: Creates a subfigure occupying 40% of the text width.
- `\hspace{0.05\textwidth}`: Adds horizontal space of 5% of the textwidth between the subfigures.

This code will insert two figures side by side with captions ‘Figure A’ and ‘Figure B’ and a common caption ‘Two figures side by side’.



(a) Figure A



(b) Figure B

Figure 1.2: Two figures side by side

1.1.3 Inserting Two Figures Stacked Vertically

To insert two figures stacked vertically, you can use the `subfigure` package similarly to the previous example. Here's an example:

```
\begin{figure}[htbp]
  \centering
  \begin{subfigure}[b]{0.5\textwidth}
    \includegraphics[width=0.8\textwidth]{path/to/your/image-a.png}
    \caption{Subfigure A}
    \label{fig:subfigva}
  \end{subfigure}
  \vspace{0.5cm}
  \begin{subfigure}[b]{0.5\textwidth}
    \includegraphics[width=0.8\textwidth]{path/to/your/image-b.png}
    \caption{Subfigure B}
    \label{fig:subfigvb}
  \end{subfigure}
  \caption{Two figures stacked vertically}
  \label{fig:stacked}
\end{figure}
```

Explanation:

- `\vspace{0.5cm}`: Adds vertical space (of 0.5cm) between the subfigures.

This will insert two figures stacked vertically with captions ‘Figure A’ and ‘Figure B’ and a common caption ‘Two figures stacked vertically’. You can reference each subfigure by itself by using its labels or you can reference both figures at once by using their common label. In Fig. 1.3a, you can see the EALS logo in color, while Fig. 1.3b shows the EALS logo in black. Both figures are included in Fig. 1.3.



(a) Subfigure A



(b) Subfigure B

Figure 1.3: Two figures stacked vertically

1.1.4 Including List of Figures

To create a list of figures use the `\listoffigures` command. The caption of each labeled figure will be used to generate this list.

1.1.5 Macros

To make the writing of equations faster and easier, this template provides several macros which are shortened versions of frequently used mathematical LaTeX functions. The definition of these macros can be found in the `macros.tex` file. Feel free to add your own macros for e.g. defining variables that are used multiple times throughout the document.

`\mrm{}`

To avoid text in a mathematical env to become *cursive*, use the macro-command `\mrm{}` which we provide in this template (see `macros.tex`).

Without `\mrm{}`:

$$V_{pp} = 6V$$

With `\mrm{}`:

$$V_{\mathrm{pp}} = 6\mathrm{V}$$

This should be remembered because only variables should be written in cursive. In this case, neither the subscript "pp" or the unit V is a variable, so it shouldn't be cursive. This macro is a short version of the LaTeX function `\mathremove{}`.

`\diff`

When writing a derivation we differentiate between a full derivation and a partial derivation. The full derivation is denoted with the letter d. This can be written in a math environment by using the macro `\diff`.

For example:

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{\mathrm{d}s}{\mathrm{d}t}$$

The partial derivation is denoted with the symbol ∂ . by using the LaTeX function `\partial`.

For example:

$$\frac{\partial Q}{\partial t} = \frac{\partial s}{\partial t}$$

`\RE`, `\IM`

To denote the real-valued part of a complex-valued variable, the symbol $\Re$ is used. This can be written in a math environment by using the macro `\RE`. To denote the imaginary-valued part of a complex-valued variable, the symbol $\Im$ is used. This can be written in a math environment by using the macro `\IM`.

Example:

$$|z| = \sqrt{\Re(z)^2 + \Im(z)^2}$$

\J

The imaginary unit is either denoted by the symbol j or i . In electrical engineering, j is more common. Because j is not a variable, it is not written in cursive. The macro `/J` enables us to use j in a mathematical environment.

Example:

$$z = a + jb$$

\T

A transposed vector is denoted with the letter T in superscript. To write this in a mathematical environment the macro `\T` can be used.

Example:

$$(A + B)^T = A^T + B^T$$

\COS,\SIN

To add $\cos$ and $\sin$ with large brackets in a mathematical environment, we can use the `\COS{}` or `\SIN{}` macro.

$$\cos\left(\frac{\omega}{\omega_0}\right)$$

1.2 Glossary

A glossary is a section at the end of a written work that defines confusing, technical, or advanced words. It is not mandatory to include a glossary in your thesis, but if you want to include one, here is how.

First, you create a new `tex`-file in which you write your glossary entries. In LaTeX you can define a glossary entry the following way:

```
\newglossaryentry{latex}{  
  name=LaTeX,  
  description={A document preparation system for high-quality typesetting}  
}
```

Explanation:

- **label**: The first parameter is the label of this term and is used to reference it within the document with the `gls` command. In the example from above the label is ‘`latex`’.
- **name**: The name parameter includes the word to be defined, in this case, ‘`LaTeX`’.
- **description**: Includes the definition of the current term.

You can also define acronyms you use in your thesis. To do this, you open the `tex`-file containing your glossary entries and then define your acronym the following way:

```
\newacronym{rms}{RMS}{Root Mean Squared}
```

The structure is the same as of the `\newglossaryentry`. Here, the **label** is `rms`, the **name** is `RMS` and the **description** is `Root Mean Squared`.

To include the `glossary.tex` file in your thesis, you use the command `\input{glossary_filename}`. Here, you have to replace the `glossary_filename` parameter with the actual filename of the `tex`-file containing your glossary. This is already defined in the `main.tex` file in line 56.

To print your glossary you have to use the command `\printglossary` at the position in your thesis at which you want the glossary to be. We inserted it at the beginning of the thesis (before the table of contents).

If you have acronyms that you want to include, you use an additional keyword with the previous command: `\printglossary[type=\acronymtype]`.

1.2.1 Use Glossary entries in the text

After you have defined the terms, to use them while you are typing your LaTeX file use one of the commands described below:

- `\gls{ }`: This prints the term in lowercase.
Example: `\gls{maths}` prints ‘`mathematics`’ when used.
- `\Gls{ }`: This prints the term lowercase with the first letter printed in uppercase.
Example: `\Gls{maths}` prints ‘`Mathematics`’.
- `\glspl{ }`: This prints the term in lowercase and puts it in its plural form.
Example: `\glspl{formula}` prints ‘`formulas`’.

- `\Glspl{ }`: This prints the term with the first letter in uppercase and puts it in its plural form.

Example: `\Glspl{formula}` prints ‘Formulas’.

Including your terms in your text, using the commands from above may look something like this:

In this thesis, we will use LaTeX to prepare a high-quality document. The LaTeX typesetting markup language is especially suitable for documents that include mathematics. Formulas are rendered easily once one gets used to the commands.

1.3 Model Predictive Control

1.3.1 Comparison to classical control

Using classical field-oriented PI current control, this is solved by splitting the system up into two linear, decoupled subsystems and two coupling terms (with the induced voltage acting affine to one of the).

$$\begin{aligned}
 \dot{i}_{Sd} &= \underbrace{\frac{1}{L_d} u_{Sd} - \frac{R_S}{L_d} i_{Sd}}_{\Sigma_{d,\text{decoupled}}} + \underbrace{\frac{L_q}{L_d} i_{Sq} \omega}_{\Sigma_{d,\text{coupled}}} \\
 \dot{i}_{Sq} &= \underbrace{\frac{1}{L_q} u_{Sq} - \frac{R_S}{L_q} i_{Sq}}_{\Sigma_{q,\text{decoupled}}} - \underbrace{\frac{L_d}{L_q} i_{Sd} \omega}_{\Sigma_{q,\text{coupled}}} - \underbrace{\frac{1}{L_q} \psi_{PM} \omega}_{\Sigma_{q,\text{affine}}}
 \end{aligned}$$

The linear systems $\Sigma_{d,\text{decoupled}}$ and $\Sigma_{q,\text{decoupled}}$ are controlled using two independent PI controllers. The remaining terms are compensated by a feed-forward system.

1.4 2 Level Inverter

2

Modelling

2.1 Interior Magnet PMSM

This leads to the to the continuous time state space model of Equation (2.1).

$$\frac{d}{dt} \begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix} = \begin{bmatrix} -\frac{R_S}{L_d} & \frac{L_q}{L_d} \omega_e(t) \\ -\frac{L_d}{L_q} \omega_e(t) & -\frac{R_S}{L_q} \end{bmatrix} \begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix} \begin{bmatrix} v_{Sd} \\ v_{Sq} \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{\psi_{PM}}{L_q} \omega_e(t) \end{bmatrix} \quad (2.1)$$

2.2 Coupling Dynamics

$$\begin{aligned} J_1 \frac{d(\omega_{m1})}{dt} &= T_e - c \cdot \Delta\varphi - k \cdot \Delta\omega_m \\ J_2 \frac{d(\omega_{m2})}{dt} &= T_l + c \cdot \Delta\varphi + k \cdot \Delta\omega_m \\ \frac{d(\varphi_{m1})}{dt} &= \omega_{m1} \\ \frac{d(\varphi_{m2})}{dt} &= \omega_{m2} \end{aligned} \quad (2.2)$$

Model Predictive Current Control

3.1 Prediction Model

Only Predict current dynamics, dont use coupling in prediction Caramaot OaC book zitieren.

3.1.1 Choice of adequate State Space Model

As derived in Section 2.1, the model of the interior magnet PMSM is given by Equation (3.1).

$$\underbrace{\frac{d}{dt} \begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix}}_{\dot{\mathbf{x}}_c} = \underbrace{\begin{bmatrix} -\frac{R_s}{L_d} & \frac{L_q}{L_d} \omega(t) \\ -\frac{L_d}{L_q} \omega(t) & -\frac{R_s}{L_q} \end{bmatrix}}_{\mathbf{A}_c(\omega(t))} \underbrace{\begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix}}_{\mathbf{x}_c} + \underbrace{\begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix}}_{\mathbf{B}_c} \underbrace{\begin{bmatrix} u_{Sd} \\ u_{Sq} \end{bmatrix}}_{\mathbf{u}_c} + \underbrace{\begin{bmatrix} 0 \\ -\frac{\psi_{PM}}{L_q} \omega(t) \end{bmatrix}}_{\mathbf{E}_c(\omega(t))} \quad (3.1)$$

For the ease of notation, $\omega = \omega_e = \omega_{m1}p$ refers to the electrical rotor speed for the rest of this thesis. Writing the nonlinear continuous time dynamic system in established control theory notation leads to the following notation:

$$\begin{aligned} \dot{\mathbf{x}}_c &= \mathbf{f}(\mathbf{x}_c, \mathbf{u}_c, \omega) \\ \dot{\mathbf{x}}_c &= \mathbf{A}_c(\omega(t))\mathbf{x}_c + \mathbf{B}_c\mathbf{u}_c + \mathbf{E}_c(\omega(t)) \end{aligned} \quad (3.2)$$

For a variety of applications it is beneficial to work with linear models, since this allows for the use of broad framework of control schemes. However, the affine term regarding the induced voltage $\mathbf{E}_c(\omega(t))$ is non-negligable for the current dynamics $\dot{\mathbf{x}}$ at every operating point except standstill or locked rotor operation.

As mentioned in Section 1.3.1, using model predictive control, one can directly use a MIMO formulation, leading to a direct consideration the D/Q axis coupling. The question remaining is whether a discretized model of Equation (3.1) shall be directly used or a step-wise linearization approach around an operating point shall be employed.

For the linearization, one would define an operating point as

$$\mathbf{x}_c = \mathbf{x}_{c,OP} + \delta \mathbf{x}_c, \quad \mathbf{u}_c = \mathbf{u}_{c,OP} + \delta \mathbf{u}_c, \quad \omega = \omega_{OP} + \delta \omega.$$

And approximate the dynamics (3.2) using a first-order Taylor approximation around an operating point $OP = (\mathbf{x}_{c,OP}, \mathbf{u}_{c,OP}, \omega_{OP})$

$$\frac{d\delta \mathbf{x}_c}{dt} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}_c} \right|_{OP} \delta \mathbf{x}_c + \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}_c} \right|_{OP} \delta \mathbf{u}_c + \left. \frac{\partial \mathbf{f}}{\partial \omega} \right|_{OP} \delta \omega.$$

Analyzing the nonlinear system (3.1), it can be seen that the only non-linearity stems from the induced voltage $\mathbf{E}_c(\omega(t))$, which is proportional to the scheduling parameter ω . As shown in Section 2, the dynamics of the mechanical system are slower than the electrical dynamics, assuming $\omega(t)$ is slowly varying relative to the sampling period T_S . Therefore, introducing step-wise linearization only leads to increased complexity without reducing the computational effort.

Thus, it was decided to use the affine prediction model of Equation (3.1) directly. This approach is also typically used in literature [3].

3.1.2 Discretization

Due to its simplicity, Forward Euler discretization shall be considered. This approach conserves stability, when the sampling time T_S is sufficiently small. The mapping of continuous time eigenvalues to discrete time eigenvalues $\underline{s}_i = a + jb \rightarrow \underline{z}_i$ is described as

$$\underline{z}_i = 1 + T_S \underline{s}_i$$

For $\underline{z}_i$ to be within the unit circle

$$\|\underline{z}_i\| = \|1 + T_S \underline{s}_i\| = 1 + 2T_S^2 a + T_S a^2 + T_S b^2 < 1 \iff T_S^2(a^2 + b^2) < -2T_S a$$

must hold. So, for positive T_S , this can be rewritten as

$$T_S \|\underline{s}_i\|^2 < -2\Re\{\underline{s}_i\} \iff T_S < \frac{-2\Re\{\underline{s}_i\}}{\|\underline{s}_i\|^2}$$

With the eigenvalue analysis of Section 2 and $T_S \leq 200 \mu s$ within the scope of this thesis, applying the Forward Euler approach will lead to stable discretization for the model of

the PMSM. The discretized state space model is then obtained by:

$$\begin{aligned}\mathbf{A}(\omega[k]) &= \mathbf{I} + T_S \mathbf{A}_c(\omega(kT_S)) \quad \mathbf{B} = T_S \mathbf{B}_c \\ \mathbf{E}(\omega[k]) &= \mathbf{E}_c(\omega(kT_S))\end{aligned}$$

In literature the actuating signal is often split up as $\mathbf{u}[k] = \mathbf{u}[k-1] + \Delta\mathbf{u}[k]$. This offers benefits as it eases the stability analysis or improves some aspects of robustness as shown in Section 3.7 (Page 31). So, the discrete time-model for prediction is given by Equation (3.3).

$$\begin{aligned}\mathbf{x}[k+1] &= \mathbf{A}(\omega[k])\mathbf{x}[k] + \mathbf{B}(\mathbf{u}[k-1] + \Delta\mathbf{u}[k]) + \mathbf{E}(\omega[k]) \\ \mathbf{x}[k+1] &= \begin{bmatrix} 1 - \frac{T_s R_S}{L_d} & T_s \frac{L_q}{L_d} \omega[k] \\ -T_s \frac{L_d}{L_q} \omega[k] & 1 - \frac{T_s R_S}{L_q} \end{bmatrix} \mathbf{x}[k] \\ &\quad + \begin{bmatrix} \frac{T_s}{L_d} & 0 \\ 0 & \frac{T_s}{L_q} \end{bmatrix} (\mathbf{u}[k-1] + \Delta\mathbf{u}[k]) + \begin{bmatrix} 0 \\ -\frac{T_s \psi_{PM}}{L_q} \omega[k] \end{bmatrix}\end{aligned}\tag{3.3}$$

3.1.3 Prediction Horizon

In model predictive one distinguishes between the prediction horizon N_P , which describes how many discrete steps in time the dynamics are predicted and the control horizon N_C , which is the amount of optimal actuating signal are computed at every step. With $N_C \leq N_P$ Within the scope of this thesis, typically field-oriented current control sampling times of $T_S \in [5, 20]$ kHz are considered. To get an optimal trade off between model fault tolerance, tracking performance and computational effort a typical prediction horizon of $N_P \leq 3$ with $N_P = N_C$ are considered.

To obtain a prediction for $\mathbf{x}[k+l]$ $l \in \mathbb{N} : l \leq N_P$, Equation (3.3) is iterated one step ahead.

$$\hat{\mathbf{x}}[k+2] = \mathbf{A}(\hat{\omega}[k+1])\hat{\mathbf{x}}[k+1] + \mathbf{B}(\mathbf{u}[k] + \Delta\mathbf{u}[k+1]) + \mathbf{E}(\hat{\omega}[k+1])$$

Using Relation (3.3) for the estimation $\hat{\mathbf{x}}[k+1]$, $\mathbf{u}[k] = \mathbf{u}[k-1] + \Delta\mathbf{u}[k]$ and assuming $\hat{\omega}[k+l] = \omega[k]$, leads to

$$\begin{aligned}\hat{\mathbf{x}}[k+2] &= \mathbf{A}(\omega[k]) (\mathbf{A}(\omega[k])\mathbf{x}[k] + \mathbf{B}(\mathbf{u}[k-1] + \Delta\mathbf{u}[k]) + \mathbf{E}(\omega[k])) \\ &\quad + \mathbf{B}(\mathbf{u}[k-1] + \Delta\mathbf{u}[k] + \Delta\mathbf{u}[k+1]) + \mathbf{E}(\omega[k]) \\ \hat{\mathbf{x}}[k+2] &= (\mathbf{A}(\omega[k]))^2 \mathbf{x}[k] + (\mathbf{A}(\omega[k]) + \mathbf{I}) \mathbf{B} \mathbf{u}[k-1] \\ &\quad + \mathbf{B} (\mathbf{A}(\omega[k]) + \mathbf{I}) \Delta\mathbf{u}[k] + \mathbf{B} \Delta\mathbf{u}[k+1] + (\mathbf{A}(\omega[k]) + \mathbf{I}) \mathbf{E}(\omega[k])\end{aligned}$$

This continues up to $l = N_P$.

$$\begin{aligned}\hat{\mathbf{x}}[k + N_P] &= (\mathbf{A}(\omega[k]))^{N_P} \mathbf{x}[k] + \left(\sum_{f=0}^{N_P-1} (\mathbf{A}(\omega[k]))^f \right) \mathbf{B} \mathbf{u}[k-1] \\ &+ \sum_{h=0}^{N_P-1} \left(\sum_{f=0}^{N_P-1-h} (\mathbf{A}(\omega[k]))^f \right) \mathbf{B} \Delta \mathbf{u}[k+h] \\ &+ \left(\sum_{f=0}^{N_P-1} (\mathbf{A}(\omega[k]))^f \right) \mathbf{E}(\omega[k])\end{aligned}$$

For the nominal system (3.3), the output is equal to the current states $y[k] = x[k]$, but for other formulations with $\mathbf{C} \neq \mathbf{I}$ it is necessary to multiply the state description from the left by $\mathbf{C}$ to obtain an estimation for the output $\hat{\mathbf{y}}[k+l]$. Doing that and using vector notations for all $l \in [1, N_P]$ of $\hat{\mathbf{y}}[k+l]$, leads to

$$\begin{aligned}\underbrace{\begin{bmatrix} \hat{\mathbf{y}}[k+1] \\ \hat{\mathbf{y}}[k+2] \\ \vdots \\ \hat{\mathbf{y}}[k+N_P] \end{bmatrix}}_{\hat{\mathbf{y}}[k+1]} &= \underbrace{\begin{bmatrix} \mathbf{C} \mathbf{A}(\omega[k]) \\ \mathbf{C} (\mathbf{A}(\omega[k]))^2 \\ \vdots \\ \mathbf{C} (\mathbf{A}(\omega[k]))^{N_P-1} \end{bmatrix}}_{\mathbf{P}_x} \mathbf{x}[k] + \underbrace{\begin{bmatrix} \mathbf{C} \mathbf{B} \\ \mathbf{C} (\mathbf{A}(\omega[k]) + \mathbf{I}) \mathbf{B} \\ \vdots \\ \mathbf{C} \left(\sum_{l=0}^{N_P-1} (\mathbf{A}(\omega[k]))^l \right) \mathbf{B} \end{bmatrix}}_{\mathbf{P}_u} \mathbf{u}[k-1] + \\ &\underbrace{\begin{bmatrix} \mathbf{C} \mathbf{B} & \mathbf{0}_2 & \cdots & \mathbf{0}_2 \\ \mathbf{C} (\mathbf{A}(\omega[k]) + \mathbf{I}) \mathbf{B} & \mathbf{C} \mathbf{B} & \cdots & \mathbf{0}_2 \\ \mathbf{C} \left((\mathbf{A}(\omega[k]))^2 + \mathbf{A}(\omega[k]) + \mathbf{I} \right) \mathbf{B} & \mathbf{C} (\mathbf{A}(\omega[k]) + \mathbf{I}) \mathbf{B} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \mathbf{0}_2 \\ \mathbf{C} \left(\sum_{l=0}^{N_P-1} (\mathbf{A}(\omega[k]))^l \right) \mathbf{B} & \mathbf{C} \left(\sum_{l=0}^{N_P-2} (\mathbf{A}(\omega[k]))^l \right) \mathbf{B} & \cdots & \mathbf{C} \mathbf{B} \end{bmatrix}}_{\mathbf{P}_{\Delta u}} \underbrace{\begin{bmatrix} \Delta \mathbf{u}[k] \\ \Delta \mathbf{u}[k+1] \\ \vdots \\ \Delta \mathbf{u}[k+N_P-1] \end{bmatrix}}_{\Delta \mathbf{u}[k]} \\ &+ \underbrace{\begin{bmatrix} \mathbf{C} \\ \mathbf{C} (\mathbf{A}(\omega[k]) + \mathbf{I}) \\ \vdots \\ \mathbf{C} \left(\sum_{l=0}^{N_P-1} (\mathbf{A}(\omega[k]))^l \right) \end{bmatrix}}_{\mathbf{P}_E} \mathbf{E}(\omega[k])\end{aligned}\tag{3.4}$$

$$\bar{\mathbf{y}}[k+1] = \underbrace{\mathbf{P}_x \bar{\mathbf{x}}[k] + \mathbf{P}_u \mathbf{u}[k-1] + \mathbf{P}_E \mathbf{E}(\omega[k])}_{=:\bar{\mathbf{g}}[k]} + \mathbf{P}_{\bar{\Delta}u} \bar{\Delta} \mathbf{u}[k] \quad (3.5)$$

With $\bar{\mathbf{g}}[k]$ being parameter variant, but independent of $\bar{\Delta} \mathbf{u}[k]$. For the implementation it is beneficial to recognize that the columns of $\mathbf{P}_{\bar{\Delta}u}$ are down-shifted columns of $\mathbf{P}_u$.

3.2 Optimization Problem

The vector of the voltage increment at the current step in time and its predictions up to the N_P th step shall be determined optimal with respect to a certain cost function J and constraints regarding the stator voltage and the predicted current response, such that the determined stator voltage can be physically applied without damaging the machine. For the actuating signal $\bar{\mathbf{u}}_S[k]$, this means it has to be bounded by the voltage hexagon and the magnitude of each $\bar{\mathbf{i}}_S[k+1]$ must not exceed the current limit I_{max} . Doing so, leads to a general description for MPC for PMSM, similar to the one used in [3].

$$\begin{aligned} \bar{\Delta} \mathbf{u}_S[k]^* &= \arg \min_{\bar{\Delta} \mathbf{u}_S} J(\mathbf{u}_S[k], \mathbf{i}_S[k], \mathbf{i}_{S,ref}[k+1]) \\ \text{s.t. } \hat{\mathbf{i}}_S[k+1] &\in \mathbb{I}_S = \{\mathbf{i}_S \in \mathbb{R}^2 : \|\mathbf{i}_S^R[k+1]\|_2 \leq I_{max}\} \\ \mathbf{u}_S^S[k] &\in \mathbb{U}_S^S = \{\mathbf{u}_S^S \in \mathbb{R}^2 : \mathbf{u}_S^S = U_{DC} \mathbf{T}_{abc \rightarrow dq} \mathbf{d}_{abc}\} \end{aligned}$$

Since the controller operates within the RRF, the relation for the voltage constraints must be transformed. To guarantee a unique global minimizer, J must be a convex function and $\mathbb{U}_S^R, \mathbb{I}_S^R$ convex sets, expressed as a set of linear inequality constraints. To use the framework of quadratic programming, the constraints must be expressed in terms of the optimization variable $\bar{\Delta} \mathbf{u}[k]$. Keeping this in mind and using index u for quantities regarding the prediction vector of the actuating signal $\bar{\mathbf{u}}_S[k]$ and i for quantities regarding the output prediction vector $\bar{\mathbf{i}}_S[k+1]$, the goal of this section is to derive a quadratic program in the form of

$$\begin{aligned} \bar{\Delta} \mathbf{u}_S^R[k]^* &= \arg \min_{\bar{\Delta} \mathbf{u}_S} \frac{1}{2} \left(\bar{\Delta} \mathbf{u}_S^R[k] \right)^T \mathbf{\Gamma} \left(\bar{\Delta} \mathbf{u}_S^R[k] \right) + \gamma^T \bar{\Delta} \mathbf{u}_S^R[k] \\ &= \arg \min_{\bar{\Delta} \mathbf{u}_S} \frac{1}{2} \|\bar{\Delta} \mathbf{u}_S^R[k]\|_{\mathbf{\Gamma}}^2 + \gamma^T \bar{\Delta} \mathbf{u}_S^R[k] \\ \text{s.t. } \underbrace{\begin{bmatrix} \mathbf{M}_u \\ \mathbf{M}_i \end{bmatrix}}_{\mathbf{M}} \bar{\Delta} \mathbf{u}_S^R[k] &\leq \underbrace{\begin{bmatrix} \boldsymbol{\mu}_u \\ \boldsymbol{\mu}_i \end{bmatrix}}_{\boldsymbol{\mu}} \end{aligned} \quad (3.6)$$

with the Hessian of the cost function being positive definite $\nabla_{\bar{\Delta}\mathbf{u}[k]}^2 J = \mathbf{\Gamma} \succ 0$ to ensure convexity.

3.2.1 Cost Function

For the cost function J , a simple LQR-style base cost is employed. With $\mathbf{Q} \in \mathbb{R}^{N_{Pq} \times N_{Pq}}$ as the diagonal weighting matrix for the tracking error and $\mathbf{R} \in \mathbb{R}^{N_{Pm} \times N_{Pm}}$ the diagonal weighting matrix for the stator voltage increment. For the application of electric drives, reference tracking is significantly more important than the magnitude of the stator voltage.

$$J = (\bar{\mathbf{i}}[k+1] - \bar{\mathbf{i}}_{ref}[k+1])^T \mathbf{Q} (\bar{\mathbf{i}}[k+1] - \bar{\mathbf{i}}_{ref}[k+1]) + (\bar{\Delta}\mathbf{u}[k])^T \mathbf{R} \bar{\Delta}\mathbf{u}[k]$$

Or with a more compact notation $\|\mathbf{x}\|_I^2 = \mathbf{x}^T \mathbf{I} \mathbf{x}$:

$$J = \|\bar{\mathbf{i}} - \bar{\mathbf{i}}_{ref}\|_Q^2 + \|\bar{\Delta}\mathbf{u}\|_R^2 \quad (3.7)$$

To express the cost J in terms of $\bar{\Delta}\mathbf{u}[k]$, the prediction for the current as output signal is expressed by the prediction based on $\Sigma : [\mathbf{A}(\omega[k]), \mathbf{B}, \mathbf{E}(\omega[k])]$ according to Equation (3.5).

$$\bar{\mathbf{i}}[k+1] = \bar{\mathbf{g}}[k+1] = \bar{\mathbf{g}}[k] + \mathbf{P}_{\bar{\Delta}u} \bar{\Delta}\mathbf{u}[k]$$

A prediction of the reference current is given as input $\bar{\mathbf{i}}_{ref}[k+1] = \bar{\mathbf{r}}[k+1]$.

$$\begin{aligned} \bar{\mathbf{i}}[k+1] - \bar{\mathbf{i}}_{ref}[k+1] &= \bar{\mathbf{g}}[k+1] - \bar{\mathbf{r}}[k+1] \\ &= \underbrace{\mathbf{P}_x \mathbf{x} + \mathbf{P}_u \mathbf{u}[-1] + \mathbf{P}_E \mathbf{E}(\omega[k])}_{\bar{\mathbf{g}}[k]} - \bar{\mathbf{r}}[k+1] + \mathbf{P}_{\bar{\Delta}u} \bar{\Delta}\mathbf{u}[k] \\ &= \bar{\tilde{\mathbf{e}}}[k+1] + \mathbf{P}_{\bar{\Delta}u} \bar{\Delta}\mathbf{u}[k] \end{aligned}$$

Here, $\bar{\tilde{\mathbf{e}}}[k+1] = \bar{\mathbf{g}}[k] - \bar{\mathbf{r}}[k+1]$ is defined as the prediction of the quasi-tracking error. This is a summation of the prediction terms independent of $\bar{\Delta}\mathbf{u}[k]$. The separation is done in order to use $\bar{\Delta}\mathbf{u}[k]$ as the vector of optimization variables. Inserting this into (3.7), leads to:

$$J = (\bar{\tilde{\mathbf{e}}}[k+1] + \mathbf{P}_{\bar{\Delta}u} \bar{\Delta}\mathbf{u}[k])^T \mathbf{Q} (\bar{\tilde{\mathbf{e}}}[k+1] + \mathbf{P}_{\bar{\Delta}u} \bar{\Delta}\mathbf{u}[k]) + (\bar{\Delta}\mathbf{u}[k])^T \mathbf{R} \bar{\Delta}\mathbf{u}[k]$$

Skipping the time indices k for the quasi-error prediction, this is expanded to

$$\begin{aligned}
 J &= \tilde{\mathbf{e}}^T \mathbf{Q} \tilde{\mathbf{e}} + 2 (\bar{\Delta} \mathbf{u}[k])^T \mathbf{P}_{\Delta u}^T \mathbf{Q} \tilde{\mathbf{e}} \\
 &\quad + (\bar{\Delta} \mathbf{u}[k])^T \mathbf{P}_{\Delta u}^T \mathbf{Q} \mathbf{P}_{\Delta u} \bar{\Delta} \mathbf{u}[k] + (\bar{\Delta} \mathbf{u}[k])^T \mathbf{R} \bar{\Delta} \mathbf{u}[k] \\
 &= (\bar{\Delta} \mathbf{u}[k])^T (\mathbf{P}_{\Delta u}^T \mathbf{Q} \mathbf{P}_{\Delta u} + \mathbf{R}) \bar{\Delta} \mathbf{u}[k] + 2 \tilde{\mathbf{e}}^T \mathbf{Q} \mathbf{P}_{\Delta u} \bar{\Delta} \mathbf{u}[k] \\
 &\quad + \tilde{\mathbf{e}}^T \mathbf{Q} \tilde{\mathbf{e}}.
 \end{aligned}$$

Due to $\tilde{\mathbf{e}}^T \mathbf{Q} \tilde{\mathbf{e}}$ being constant in relation to the optimization vector, it can be ignored. Using the aforementioned compact notation and coefficient comparison for the matrices of the Quadratic Program (3.6) can be conducted.

$$\begin{aligned}
 J &= \|\bar{\Delta} \mathbf{u}[k]\|_{(\mathbf{P}_{\Delta u})^T \mathbf{Q} \mathbf{P}_{\Delta u} + \mathbf{R}}^2 & + & \quad 2 (\tilde{\mathbf{e}}[k+1])^T \mathbf{Q} \mathbf{P}_{\Delta u} \bar{\Delta} \mathbf{u}[k] \\
 &\stackrel{!}{=} \frac{1}{2} \|\bar{\Delta} \mathbf{u}[k]\|_{\mathbf{\Gamma}}^2 & + & \quad \gamma^T \bar{\Delta} \mathbf{u}[k]
 \end{aligned}$$

Keeping the factor 1/2 for the quadratic term in mind, the matrices $\mathbf{\Gamma}, \gamma$ are defined as

$$\mathbf{\Gamma} = 2 \left((\mathbf{P}_{\Delta u})^T \mathbf{Q} \mathbf{P}_{\Delta u} + \mathbf{R} \right) \quad \gamma^T = 2 (\tilde{\mathbf{e}}[k+1])^T \mathbf{Q} \mathbf{P}_{\Delta u}$$

To ensure convexity, $\mathbf{Q}, \mathbf{R}$ must be chosen such that $\mathbf{\Gamma} \succ 0$.

3.2.2 Voltage Constraints

As mentioned in Section 1.4, the stator voltage vector $\mathbf{u}_S^S$ is subject to the constraints of the voltage hexagon spanned by the 6 active vectors. These constraints can be formulated as a set of linear inequality constraints. For example, the upper flat border of the hexagon is described by

$$u_\beta \leq \frac{U_{DC}}{\sqrt{3}} \tag{3.8}$$

and the upper right border, with a slope of $-\pi/3$ by

$$u_\beta \leq -\sqrt{3} u_\alpha + \frac{2 U_{DC}}{\sqrt{3}} \tag{3.9}$$

Both of the inequalities (3.8) and (3.9) describe convex sets. Continuing this construction of the voltage hexagon for all six borders, leads to a sixfold convex intersection of half-planes as in [4], which describes the voltage hexagon. Combining those six relations leads to the set of Linear Inequalities in SRF (3.10).

$$\underbrace{\begin{bmatrix} \sqrt{3} & 1 \\ 0 & 1 \\ -\sqrt{3} & 1 \\ -\sqrt{3} & -1 \\ 0 & -1 \\ \sqrt{3} & -1 \end{bmatrix}}_{\mathbf{N}_u^S} \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} \leq \frac{U_{DC}}{\sqrt{3}} \underbrace{\begin{bmatrix} 2 \\ 1 \\ 2 \\ 2 \\ 1 \\ 2 \end{bmatrix}}_{\boldsymbol{\nu}_u^S} \quad (3.10)$$

Using Equation (3.10), the voltage constraints in the stator reference frame can be compactly written down as:

$$\mathbf{N}_u^S \mathbf{u}_S^S[k] \leq \boldsymbol{\nu}_u^S$$

However, the optimization problem shall be defined with respect to the the rotor reference frame in terms of the incremental stator voltage predictions. To do so, first, the rotation from the RRF is applied to $\mathbf{u}_S^R[k] = \boldsymbol{\Delta} \mathbf{u}_S^R[k] + \mathbf{u}_S^R[k-1]$ for one voltage vector.

$$\mathbf{N}_u^S \mathbf{T}_{dq \rightarrow \alpha\beta}(\varphi[k]) (\boldsymbol{\Delta} \mathbf{u}_S^R[k] + \mathbf{u}_S^R[k-1]) \leq \boldsymbol{\nu}_u^S$$

The next step is to extend that formulation to the incremental prediction vector $\boldsymbol{\Delta} \mathbf{u}_S^R[k] \rightarrow \bar{\boldsymbol{\Delta}} \mathbf{u}_S^R[k]$. To calculate $\mathbf{u}_S^R[k+1] \dots \mathbf{u}_S^R[k+N_P-1]$ from $\mathbf{u}_S^R[k-1]$ and $\boldsymbol{\Delta} \mathbf{u}_S^R[k+l]$, all the $\boldsymbol{\Delta} \mathbf{u}_S^R[k+l]$ with $l \leq N_P-1$ must be accumulated. Thus, the prediction vector in RRF is multiplied by the lower block-triangular matrix $\mathbf{P}_T$ with $\mathbf{I}_2$ as block elements.

$$\bar{\mathbf{u}}[k] = \mathbf{P}_T \bar{\boldsymbol{\Delta}} \mathbf{u}[k] + (\mathbf{1}_{N_P} \otimes \mathbf{I}_2) \mathbf{u}[k-1]$$

With $(\mathbf{1}_{N_P} \otimes \mathbf{I}_2)$ being the Kronecker product, describing N_P stacked identity matrices. Putting everything together leads to:

$$\bar{\mathbf{N}}_u^S \bar{\mathbf{T}}_{dq \rightarrow \alpha\beta}(\bar{\varphi}[k]) \left(\mathbf{P}_T \bar{\boldsymbol{\Delta}} \mathbf{u}_S^R[k] + (\mathbf{1}_{N_P} \otimes \mathbf{I}_2) \mathbf{u}_S^R[k-1] \right) \leq \bar{\boldsymbol{\nu}}_u^S \quad (3.11)$$

Or in more detail:

$$\underbrace{\begin{bmatrix} N_u^S & \mathbf{0}^{6 \times 2} & \dots & \mathbf{0}^{6 \times 2} \\ \mathbf{0}^{6 \times 2} & N_u^S & \dots & \mathbf{0}^{6 \times 2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}^{6 \times 2} & \mathbf{0}^{6 \times 2} & \dots & N_u^S \end{bmatrix}}_{\bar{N}_u^S} \underbrace{\begin{bmatrix} T_{dq \rightarrow \alpha\beta}(\varphi[k]) & \mathbf{0}_2 & \dots & \mathbf{0}_2 \\ \mathbf{0}_2 & T_{dq\alpha\beta}(\varphi[k+1]) & \dots & \mathbf{0}_2 \\ \vdots & \vdots & \ddots & \mathbf{0}_2 \\ \mathbf{0}_2 & \mathbf{0}_2 & \dots & T_{dq\alpha\beta}(\varphi[k+N_P-1]) \end{bmatrix}}_{\bar{T}_{dq \rightarrow \alpha\beta}(\bar{\varphi}[k])} \left(\underbrace{\begin{bmatrix} I_2 & \mathbf{0}_2 & \mathbf{0}_2 & \dots & \mathbf{0}_2 \\ I_2 & I_2 & \mathbf{0}_2 & \dots & \mathbf{0}_2 \\ \vdots & \vdots & \ddots & & \mathbf{0}_2 \\ I_2 & I_2 & I_2 & \dots & I_2 \end{bmatrix}}_{P_T} \underbrace{\begin{bmatrix} \Delta \mathbf{u}[k] \\ \Delta \mathbf{u}[k+1] \\ \vdots \\ \Delta \mathbf{u}[k+N_P-1] \end{bmatrix}}_{\Delta \mathbf{u}_S^R[k]} + \begin{bmatrix} I_2 \\ I_2 \\ \vdots \\ I_2 \end{bmatrix} \mathbf{u}_S^R[k-1] \right) \leq \begin{bmatrix} \nu_u^S \\ \nu_u^S \\ \vdots \\ \nu_u^S \end{bmatrix}$$

As a final step, to arrive at an LMI in terms of $\Delta \mathbf{u}[k]$, Equation (3.11) must be rearranged to:

$$\bar{N}_u^S \bar{T}_{dq \rightarrow \alpha\beta}(\bar{\varphi}[k]) P_T \Delta \mathbf{u}_S^R[k] \leq \bar{\nu}_u^S - \bar{N}_u^S \bar{T}_{dq \rightarrow \alpha\beta}(\bar{\varphi}[k]) (\mathbf{1}_{N_P} \otimes I_2) \mathbf{u}_S^R[k-1]$$

$T_{dq \rightarrow \alpha\beta}(\varphi[k])$ is part of the SO_2 group (Special-Orthogonal in the plane $\mathbb{R}^2$) [5] [6]. Thus, using the assumption $\omega[k] = \omega[k+l] : l \in [1, N_P]$ for $\varphi[k+l] = \varphi[k] + l \cdot \omega[k] T_S$ the transformation matrices for future steps can be approximated by the following product:

$$T_{dq\alpha\beta}(\varphi[k+l]) = (T_{dq \rightarrow \alpha\beta}(\omega[k] T_S))^l T_{dq \rightarrow \alpha\beta}(\varphi[k])$$

To further reduce the computational burden, the assumption $\omega[k] T_S \leq \omega_{max} T_{S,max} \leq 0.2$ holds true $\forall k$ of electric drives considered within the scope of this thesis. This allows for small angle approximation.

$$T_{dq \rightarrow \alpha\beta}(\omega[k] T_S) \approx \begin{bmatrix} 1 & -\omega[k] T_S \\ \omega[k] T_S & 1 \end{bmatrix} := \tilde{T}_{dq\alpha\beta}(\omega[k] T_S)$$

With $\tilde{\bar{T}}_{dq\alpha\beta}(\bar{\varphi}[k])$ being an block diagonal matrix of elements $\tilde{T}_{dq\alpha\beta}(\omega[k] T_S)$, the following computational-effective implementation can be written down.

$$\underbrace{\bar{N}_u^S \tilde{\bar{T}}_{dq\alpha\beta}(\bar{\varphi}[k]) P_T}_{M_u} \Delta \mathbf{u}_S^R[k] \leq \underbrace{\bar{\nu}_u^S - \bar{N}_u^S \tilde{\bar{T}}_{dq\alpha\beta}(\bar{\varphi}[k]) (\mathbf{1}_{N_P} \otimes I_2) \mathbf{u}_S^R[k-1]}_{\mu_u} \quad (3.12)$$

3.2.3 Current Constraints

To ensure safe operation, the current space vector must not exceed its limit I_{max} . Due to the rotational symmetry of this constraint, the formulation is equivalent for SRF and RRF.

$$\|\mathbf{i}_S^S\|_2 \leq I_{max} \iff \|\mathbf{i}_S^R\|_2 \leq I_{max} \quad (3.13)$$

In order to allow for short-term violation of this constraint, similar to I^2t safety protections, soft constraints are introduced in Section 3.5.3.

The Constraint 3.13 describes a disk on the complex plane. In a lot MPC literature this is approximated by a polygon, to allow for a constraint description as a finite set of linear inequality constraints. However, this usually leads to a high amount of inequalities [7]. One approach to reduce that, it so introduce parameter-varying constraints, as in [3].

Parameter-Invariant Constraints

Constraint (3.13) was approximated by an octagon. The octagon is oriented in a way, such that the $\pm i_q$ under approximated constraints are exactly equal I_{max} . Analog to the procedure described in Section 3.2.2 for the voltage hexagon, an octagon is constructed. For the full analytic derivation of the set of inequalities for the octagon, refer to Section 5.2 of [7].

$$\underbrace{\begin{bmatrix} 1 & \frac{1-c}{c} \\ \frac{1-c}{c} & 1 \\ -\frac{1-c}{c} & 1 \\ -1 & \frac{1-c}{c} \\ -1 & -\frac{1-c}{c} \\ -\frac{1-c}{c} & -1 \\ \frac{1-c}{c} & -1 \\ 1 & -\frac{1-c}{c} \end{bmatrix}}_{\mathbf{N}_i^R} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \leq \underbrace{\begin{bmatrix} I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \end{bmatrix}}_{\boldsymbol{\nu}_i^R}$$

With the auxiliary variable $c = \cos \pi/4$. The constraints regarding the stator current can be written down as:

$$\mathbf{N}_i^R \mathbf{i}_S^R[k+1] \leq \boldsymbol{\nu}_i^R$$

To arrive at a set of inequalities in terms of $\boldsymbol{\Delta u}$ for the Quadratic Program (3.6), $\mathbf{i}_S^R[k+1]$

is expressed by $\Sigma : [\mathbf{A}(\omega[k]), \mathbf{B}, \mathbf{E}(\omega[k])]$ with $\mathbf{u}_S^R[k] = \mathbf{u}_S^R[k-1] + \Delta \mathbf{u}_S^R[k]$

$$\mathbf{N}_i^R (\mathbf{A}(\omega[k]) \mathbf{i}_S^R[k] + \mathbf{B} (\mathbf{u}_S^R[k-1] + \Delta \mathbf{u}_S^R[k]) + \mathbf{E}(\omega[k])) \leq \boldsymbol{\nu}_i^R$$

Using the Prediction Equation (3.5), the constraints for the prediction vector $\bar{\Delta \mathbf{u}}[k]$ can be derived

$$\bar{\mathbf{N}}_i^R (\bar{\mathbf{g}}[k] + \mathbf{P}_{\bar{\Delta u}}(\omega[k]) \bar{\Delta \mathbf{u}}_S^R[k]) \leq \bar{\boldsymbol{\nu}}_i^R$$

or in more detail:

$$\underbrace{\begin{bmatrix} \mathbf{N}_i^R & \mathbf{0}^{8 \times 2} & \dots & \mathbf{0}^{8 \times 2} \\ \mathbf{0}^{8 \times 2} & \mathbf{N}_i^R & \dots & \mathbf{0}^{8 \times 2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}^{8 \times 2} & \mathbf{0}^{8 \times 2} & \dots & \mathbf{N}_i^R \end{bmatrix}}_{\bar{\mathbf{N}}_i^R} \left(\underbrace{\begin{bmatrix} \mathbf{g}[k] \\ \mathbf{g}[k+1] \\ \vdots \\ \mathbf{g}[k+N_P-1] \end{bmatrix}}_{\bar{\mathbf{g}}[k]} + \mathbf{P}_{\bar{\Delta u}}(\omega[k]) \underbrace{\begin{bmatrix} \Delta \mathbf{u}[k] \\ \Delta \mathbf{u}[k+1] \\ \vdots \\ \Delta \mathbf{u}[k+N_P-1] \end{bmatrix}}_{\bar{\Delta \mathbf{u}}_S^R[k]} \right) \leq \underbrace{\begin{bmatrix} \boldsymbol{\nu}_i^R \\ \boldsymbol{\nu}_i^R \\ \vdots \\ \boldsymbol{\nu}_i^R \end{bmatrix}}_{\bar{\boldsymbol{\nu}}_i^R}$$

This equation is the rearranged in terms of $\mathbf{M}_i \bar{\Delta \mathbf{u}}_S^R[k] \leq \boldsymbol{\mu}_i$:

$$\underbrace{\bar{\mathbf{N}}_i^R \mathbf{P}_{\bar{\Delta u}}}_{\mathbf{M}_i} \bar{\Delta \mathbf{u}}_S^R[k] \leq \underbrace{\bar{\boldsymbol{\nu}}_i^R - \bar{\mathbf{N}}_i^R \bar{\mathbf{g}}[k]}_{\boldsymbol{\mu}_i}$$

Reduced Order Parameter-Variant Constraints

3.2.4 Solver

3.3 MPC Formulations

3.3.1 Basic MPC

The prediction according to Equation (3.5) uses absolute states $\mathbf{x}[k]$ and differential actuating signal $\Delta \mathbf{u}[k] = \mathbf{u}[k] - \mathbf{u}[k-1]$. This enables the utilization of a commonly used MPC structure, as described in Section 2.3 of [8], which is depicted in Figure 3.1.

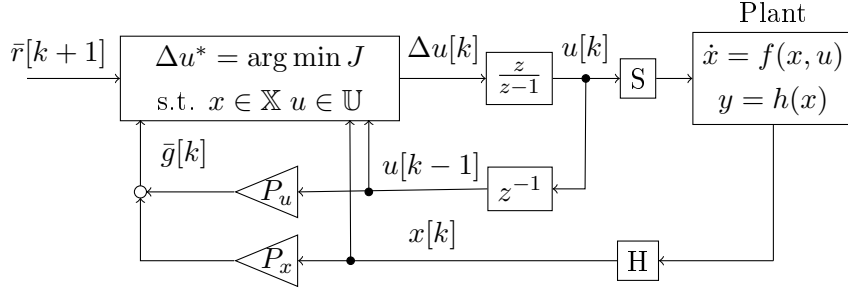


Figure 3.1: MPC structure used in [8] with the notation used in this thesis

The MPC is provided with a reference prediction

$$\bar{\mathbf{r}}[k+1] = [\mathbf{r}^T[k+1] \ \mathbf{r}^T[k+2] \ \dots \ \mathbf{r}^T[k+N_P]]^T$$

. Using current states and the previous input, the auxiliary prediction vector $\bar{\mathbf{g}}[k]$ can be computed according to Equation 3.5. Using $\tilde{\mathbf{e}}[k+1] = \bar{\mathbf{g}}[k] - \bar{\mathbf{r}}[k+1]$ to calculate the linear term γ of the quadratic program, the cost function $J = \|\Delta \mathbf{u}\|_{\Gamma}^2 + \gamma^T \Delta \bar{\mathbf{u}}[k]$ is fully determined. To determine the constraints $\mathbf{M} \Delta \bar{\mathbf{u}}[k] \leq \boldsymbol{\mu}$, measured state vector and previous input are required. The output of the Quadratic Program (3.6) is the incremental input $\Delta \bar{\mathbf{u}}[k]$, which is accumulated then to $\mathbf{u}[k]$. The interface to the plant is modeled as Sample and Hold elements.

Section 3.4 provides an analysis on why this approach does not inherently guarantee offset-free tracking, but can be extended by using the approach discussed in Section 3.5.2.

3.3.2 MPC with Embedded Integrator

A different approach is used by L. Wang [9]. Here, the state space system is augmented, such that the states and actuating signals are incremental. As it will be shown in Section 3.4, this ensures the MPC is of integrating behavior for locally unconstrained operation. In this thesis a brief overview of the derivation of the augmented state space model is given. For more details, refer to Section 1.2 of the MPC Book by Wang [9].

For discrete-time, linear time-invariant systems, without feedthrough term $\Sigma : [\mathbf{A}, \mathbf{B}, \mathbf{C}]$, the incremental states and inputs are defined as:

$$\Delta \mathbf{x}[k] = \mathbf{x}[k] - \mathbf{x}[k-1] \qquad \Delta \mathbf{u}[k] = \mathbf{u}[k] - \mathbf{u}[k-1]$$

In order to predict the absolute state or output $\mathbf{y}$, an additional state is introduced that accumulates the incremental quantities. This leads to the following augmented state-space model:

$$\begin{aligned} \begin{bmatrix} \Delta \mathbf{x}[k+1] \\ \mathbf{y}[k+1] \end{bmatrix} &= \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{CA} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x}[k] \\ \mathbf{y}[k] \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{CB} \end{bmatrix} \Delta \mathbf{u}[k] \\ \mathbf{y}[k] &= \begin{bmatrix} \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x}[k] \\ \mathbf{y}[k] \end{bmatrix} \end{aligned} \quad (3.14)$$

Applying this to the description of the affine, parameter-varying discrete time model used for the PMSM (3.3), one has to take especially care for the ω -varying terms:

$$\begin{aligned} \Sigma_{k+1} : [\mathbf{A}(\omega[k]), \mathbf{B}, \mathbf{E}(\omega[k])] - \Sigma_k : [\mathbf{A}(\omega[k-1]), \mathbf{B}, \mathbf{E}(\omega[k-1])] = \\ \mathbf{x}[k+1] - \mathbf{x}[k] = (\mathbf{A}(\omega[k]) (\mathbf{x}[k-1] + \Delta \mathbf{x}[k]) + \mathbf{B}\mathbf{u}[k] + \mathbf{E}(\omega[k])) \\ - (\mathbf{A}(\omega[k-1])\mathbf{x}[k-1] + \mathbf{B}\mathbf{u}[k-1] + \mathbf{E}(\omega[k-1])) \\ \Delta \mathbf{x}[k+1] = \mathbf{A}(\omega[k])\Delta \mathbf{x}[k] + \mathbf{B} \cdot \Delta \mathbf{u}[k] + \underbrace{\Delta \mathbf{A}(\Delta \omega[k])\mathbf{x}[k-1] + \Delta \mathbf{E}(\Delta \omega[k])}_{\mathbf{F}[k]} \end{aligned}$$

with

$$\Delta \mathbf{A}(\Delta \omega[k]) = \begin{bmatrix} \mathbf{0} & T_s \frac{L_q}{L_d} \Delta \omega[k] \\ -T_s \frac{L_d}{L_q} \Delta \omega[k] & \mathbf{0} \end{bmatrix} \quad \Delta \mathbf{E}(\Delta \omega[k]) = \begin{bmatrix} 0 \\ -\frac{T_s \psi_{PM}}{L_q} \Delta \omega[k] \end{bmatrix}.$$

Applying the incremental state accumulation as in Equation (3.14), the augmented state space system for the PMSM can be derived:

$$\begin{aligned} \underbrace{\begin{bmatrix} \Delta \mathbf{x}[k+1] \\ \mathbf{y}[k+1] \end{bmatrix}}_{\boldsymbol{\xi}[k+1]} &= \underbrace{\begin{bmatrix} \mathbf{A}(\omega[k]) & \mathbf{0} \\ \mathbf{CA}(\omega[k]) & \mathbf{I} \end{bmatrix}}_{\mathbf{A}(\omega[k])_{\Delta}} \underbrace{\begin{bmatrix} \Delta \mathbf{x}[k] \\ \mathbf{y}[k] \end{bmatrix}}_{\boldsymbol{\xi}[k]} + \underbrace{\begin{bmatrix} \mathbf{B} \\ \mathbf{CB} \end{bmatrix}}_{\mathbf{B}_{\Delta}} \Delta \mathbf{u}[k] + \underbrace{\begin{bmatrix} \mathbf{F}[k] \\ \mathbf{CF}[k] \end{bmatrix}}_{\mathbf{E}_{\Delta}[k]} \\ \mathbf{y}[k] &= \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{I} \end{bmatrix}}_{\mathbf{C}_{\Delta}} \begin{bmatrix} \Delta \mathbf{x}[k] \\ \mathbf{y}[k] \end{bmatrix} \end{aligned} \quad (3.15)$$

Apart from the ambiguity regarding $\Delta \omega$, implementing constraints with incremental quantities for $\mathbf{A}(\omega[k])$ and $\mathbf{E}(\omega[k])$ proved to introduce additional complexity.

For the implementation of this augmented system, one has to obtain the incremental state change $\Delta \mathbf{x}[k]$. This is accomplished by the filter $1 - z^{-1}$. To calculate the constraints and tracking error, $u[k - 1]$ and $x[k]$ are needed. This results into the controller structure shown in Figure 3.2.

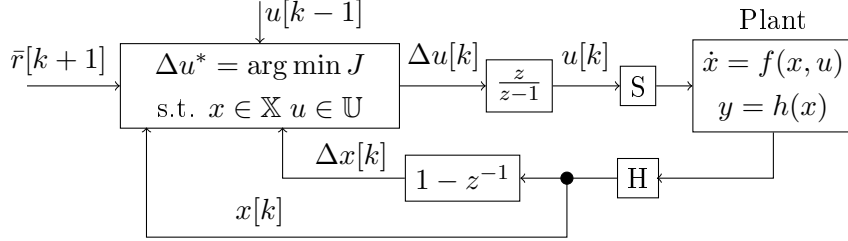


Figure 3.2: MPC structure used in [9] with the notation used in this thesis

The advantage of this approach, which introduces integrating behavior, is accompanied by the disadvantages of incremental states.

3.3.3 Hybrid Approach

Since the basic MPC formulation does not inherently guarantee offset-free tracking under output disturbances, such as model faults and the embedded integrator according to Wang hinders extensions and constraint formulations due to its incremental states, a way searched to combine the benefits of those two approaches.

To obtain an MPC using absolute states $\mathbf{x}[k] = [i_d \ i_q]^T$ and still having no direct feedback for $\mathbf{u}[k - 1]$ for the locally unconstrained controller, like in the Embedded Integrator structure of, a hybrid structure of these two approaches was designed. The result is depicted in Figure 3.3.

The idea behind this structure is to use the state description to estimate what input $\tilde{\mathbf{u}}[k - 1]$ was applied to the plant such that it leads to the measured states $x[k]$. By doing so, input disturbances $\mathbf{d}_1[k - 1] = \tilde{\mathbf{u}}[k - 1] - \mathbf{u}[k - 1]$ are considered.

$$\mathbf{x}[k] = \mathbf{A}\mathbf{x}[k - 1] + \mathbf{B} \underbrace{(\mathbf{d}_1[k - 1] + \mathbf{u}[k - 1])}_{\tilde{\mathbf{u}}[k-1]}$$

$$\tilde{\mathbf{u}}[k - 1] = \mathbf{B}^{-1}(\mathbf{x}[k] - \mathbf{A}\mathbf{x}[k - 1]).$$

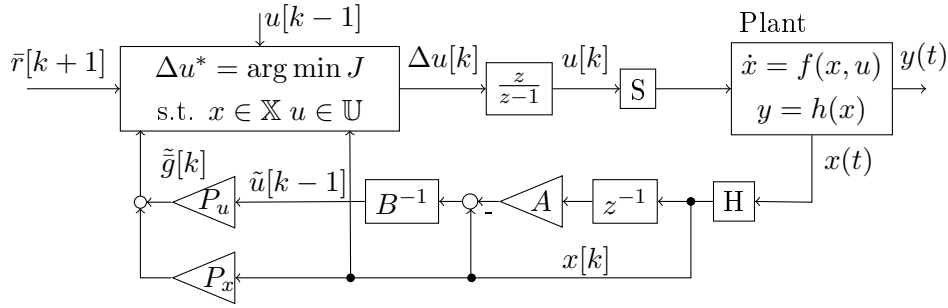


Figure 3.3: Hybrid MPC structure for LTI systems

Generally speaking, the inverse $\mathbf{B}^{-1}$ is not guaranteed to exist. However, in the case of the PMSM model used (3.3), $\mathbf{B}$ is a diagonal matrix with positive elements. With L_d, L_q being in the range of $[0.1, 10]$ mH, $\mathbf{B}$ is well conditioned. For the affine, parameter-varying prediction model $\mathbf{E}(\omega[k])$ has to be considered as well:

$$\tilde{\mathbf{u}}[k-1] = \mathbf{B}^{-1} (\mathbf{x}[k] - \mathbf{A}(\omega[k-1])\mathbf{x}[k-1] - \mathbf{E}(\omega[k-1])).$$

The dynamic behavior of this approach will be analyzed and compared to the other two formulations in the Section 3.4.

3.4 Analysis of the Controller Structure

Ziel: Integrales Verhalten und Sprungfähigkeit analysieren

3.4.1 Closed-Loop Structure with inactive Constraints

For the unconstrained case, the Quadratic Program can be solved offline by finding the stationary points of J with respect to $\bar{\Delta \mathbf{u}}[k]$:

$$\begin{aligned} \nabla_{\bar{\Delta \mathbf{u}}[k]} J &= \mathbf{\Gamma} \bar{\Delta \mathbf{u}}[k] + \boldsymbol{\gamma} = 0 \\ \bar{\Delta \mathbf{u}}^*[k] &= -\mathbf{\Gamma}^{-1} \boldsymbol{\gamma} \end{aligned}$$

With $\nabla_{\bar{\Delta \mathbf{u}}[k]}^2 J = \mathbf{\Gamma} \succ 0$ ensuring the stationary point to be a minimizer and $\mathbf{\Gamma} \succ 0$ is ensured by the choice of $\mathbf{Q}, \mathbf{R}$.

In model predictive control only the first signal of the computed optimal input prediction is applied, thus the locally unconstrained linear control law $\bar{\Delta \mathbf{u}}[k] = \mathbf{K} \bar{\mathbf{e}}[k]$ can be

$$\mathbf{u}[k] = \underbrace{\begin{bmatrix} \mathbf{I} - \mathbf{K}\mathbf{P}_u(\omega[k]) \end{bmatrix}}_{\mathbf{A}_{C,b}} \underbrace{\mathbf{u}[k-1]}_{\boldsymbol{\xi}_{C,b}[k-1]} + \underbrace{\begin{bmatrix} \mathbf{K} & -\mathbf{K}\mathbf{P}_x(\omega[k]) \end{bmatrix}}_{\mathbf{B}_{C,b}} \underbrace{\begin{bmatrix} \bar{\mathbf{r}}[k+1] \\ \mathbf{x}[k] \end{bmatrix}}_{\mathbf{v}_{C,b}[k-1]} \quad (3.19)$$

$$\mathbf{y}_{C,b}[k] = \underbrace{\mathbf{A}_{C,b}}_{\mathbf{C}_{C,b}} \boldsymbol{\xi}_{C,b}[k-1] + \underbrace{\mathbf{B}_{C,b}}_{\mathbf{D}_{C,b}} \mathbf{v}_{C,b}[k-1] = \mathbf{u}[k]. \quad (3.20)$$

The spectrum of the unconstrained Basic Controller's system matrix $\mathbf{A}_{C,b}(\omega)$, was quantitatively analyzed by plotting its eigenvalues for discrete values of $\omega \in \mathcal{W} = [-10^3, 10^3]$ rad/s. The result can be seen in Figure 3.5.

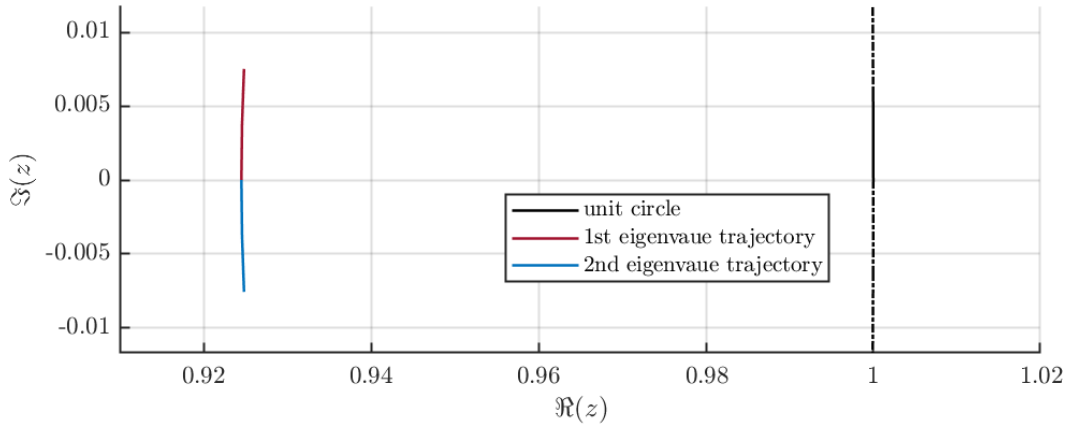


Figure 3.5: Spectrum of $\mathbf{A}_{C,b}(\omega)$ for $\omega \in \mathcal{W}$ of the unconstrained Basic MPC.

By deriving the transfer function matrix $\mathbf{v} \rightarrow \mathbf{y}$ for frozen parameter $\omega = \omega_l$, the frequency response can be analyzed.

$$\mathbf{G}_{vy}|_{\omega_l} = \mathbf{C}_{C,b}|_{\omega_l} \left(z\mathbf{I} - \mathbf{A}_{C,b}|_{\omega_l} \right)^{-1} \mathbf{B}_{C,b} + \mathbf{D}_{C,b}$$

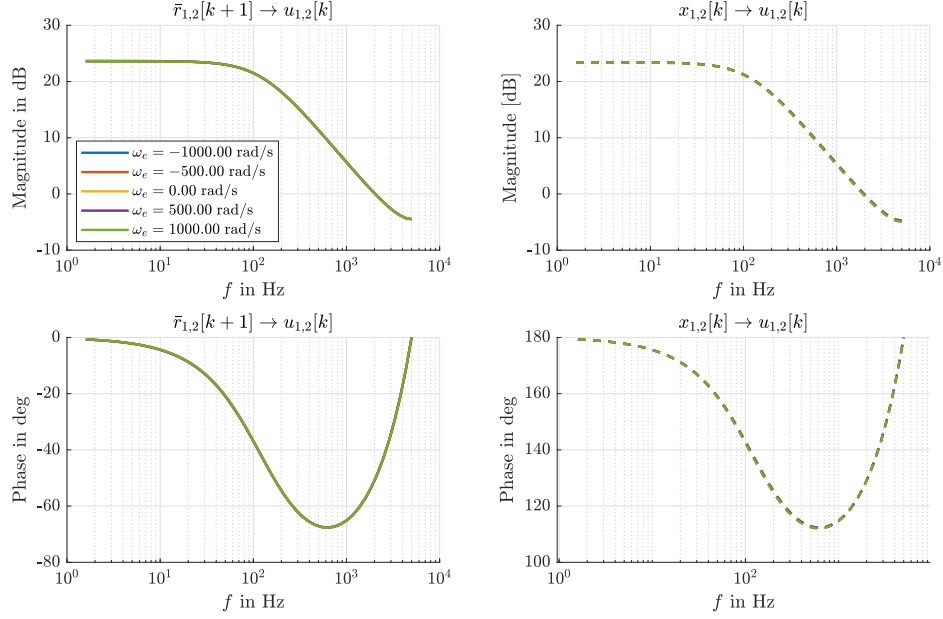


Figure 3.6: Frequency response for the unconstrained Basic MPC.

Unconstrained Hybrid MPC for LPV Systems

Again, to improve the readability of this section, the parameter dependencies ω are not explicitly stated.

The reconstructed plant input is defined by Equation (3.21).

$$\tilde{\mathbf{u}}[k-1] = \mathbf{B}^{-1}(\mathbf{x}[k] - \mathbf{A}\mathbf{x}[k-1]) \quad (3.21)$$

The control law is linear in terms of the quasi-tracking error with the optimal control gain $\mathbf{K}$ according to Equation (3.18) $\Delta \mathbf{u}[k] = \mathbf{K}\tilde{\mathbf{e}}[k]$. The accumulated plant input is then given by

$$\mathbf{u}[k] = \mathbf{u}[k-1] + \mathbf{K}\tilde{\mathbf{e}}[k] \quad (3.22)$$

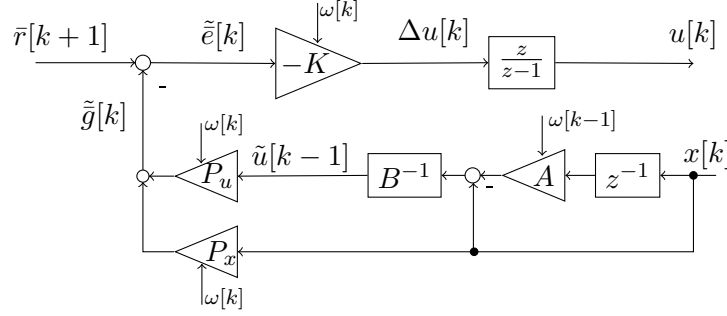


Figure 3.7: Unconstrained Hybrid MPC structure for LPV systems

Using $\tilde{e}[k] = \bar{r}[k+1] - \tilde{g}[k]$, with the definition of $\tilde{g}[k]$ according to the structure depicted in Figure 3.7, Equation (3.22) can be expanded to:

$$\begin{aligned} \mathbf{u}[k] &= \mathbf{u}[k-1] + \mathbf{K} (\bar{\mathbf{r}}[k+1] - \mathbf{P}_u (\mathbf{B}^{-1} (\mathbf{x}[k] - \mathbf{A}\mathbf{x}[k-1])) - \mathbf{P}_x \mathbf{x}[k]) \\ &= \mathbf{u}[k-1] + \mathbf{K} \mathbf{P}_u \mathbf{B}^{-1} \mathbf{x}[k-1] + \mathbf{K} \bar{\mathbf{r}}[k+1] - \mathbf{K} \mathbf{P}_u \mathbf{B}^{-1} \mathbf{x}[k] - \mathbf{K} \mathbf{P}_x \mathbf{x}[k]. \end{aligned}$$

After introducing a shift register for $\mathbf{x}[k-1]$, this can be written in matrix notation

$$\begin{bmatrix} \mathbf{u}[k] \\ \mathbf{x}[k] \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{K} \mathbf{P}_u \mathbf{B}^{-1} \mathbf{A} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u}[k-1] \\ \mathbf{x}[k-1] \end{bmatrix} + \begin{bmatrix} -\mathbf{K} & \mathbf{K} (\mathbf{P}_u \mathbf{B}^{-1} + \mathbf{P}_x) \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{r}}[k+1] \\ \mathbf{x}[k] \end{bmatrix} \quad (3.23)$$

For that system, the measured states and the reference $\mathbf{v}_{C,h} = [\bar{\mathbf{r}} \ \mathbf{x}]^T$ act as input to the controller and the stator voltage as output $\mathbf{y}_{C,h}[k] = \mathbf{u}[k]$. To derive a state space formulation the state vector is defined as the following $\boldsymbol{\xi}_{C,h}^T = [\mathbf{u} \ \mathbf{x}]$.

$$\begin{bmatrix} \mathbf{u}[k] \\ \mathbf{x}[k] \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I} & -\mathbf{K} \mathbf{P}_u(\omega[k]) \mathbf{B}^{-1} \mathbf{A}(\omega[k-1]) \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{A}_{C,h}} \begin{bmatrix} \mathbf{u}[k-1] \\ \mathbf{x}[k-1] \end{bmatrix} + \quad (3.24)$$

$$\underbrace{\begin{bmatrix} -\mathbf{K} & \mathbf{K} (\mathbf{P}_u(\omega[k]) \mathbf{B}^{-1} + \mathbf{P}_x(\omega[k])) \\ \mathbf{0} & \mathbf{I} \end{bmatrix}}_{\mathbf{B}_{C,h}} \begin{bmatrix} \bar{\mathbf{r}}[k+1] \\ \mathbf{x}[k] \end{bmatrix} \quad (3.25)$$

$$\mathbf{y}_{C,h}[k] = [\mathbf{I} \ \mathbf{0}] \cdot \mathbf{A}_{C,h} \cdot \boldsymbol{\xi}[k-1] + [\mathbf{I} \ \mathbf{0}] \cdot \mathbf{B}_{C,h} \cdot \mathbf{v}_{C,h}[k-1] = \mathbf{u}[k] \quad (3.26)$$

According to the LPV State Space Model (3.26), the system matrix $\mathbf{A}_{C,h}$ is an upper

block-triangular matrix. Thus the eigenvalues are on the main diagonal of it. Since those entries are constant, the spectrum of this system matrix is given by

$$\sigma(A_{C,h}(\omega)) = \{1, 1, 0, 0\}$$

With two discrete-time eigenvalues at one, it can be said, that this controller is of integrating behavior.

Again, the transfer function matrix $\mathbf{v} \rightarrow \mathbf{y}$ for frozen parameter $\omega = \omega_l$ is derived, so that the frequency response can be analyzed.

$$\mathbf{G}_{vy}|_{\omega_l} = \mathbf{C}_{C,h}|_{\omega_l} \left(z\mathbf{I} - \mathbf{A}_{C,h}|_{\omega_l} \right)^{-1} \mathbf{B}_{C,h} + \mathbf{D}_{C,h}$$

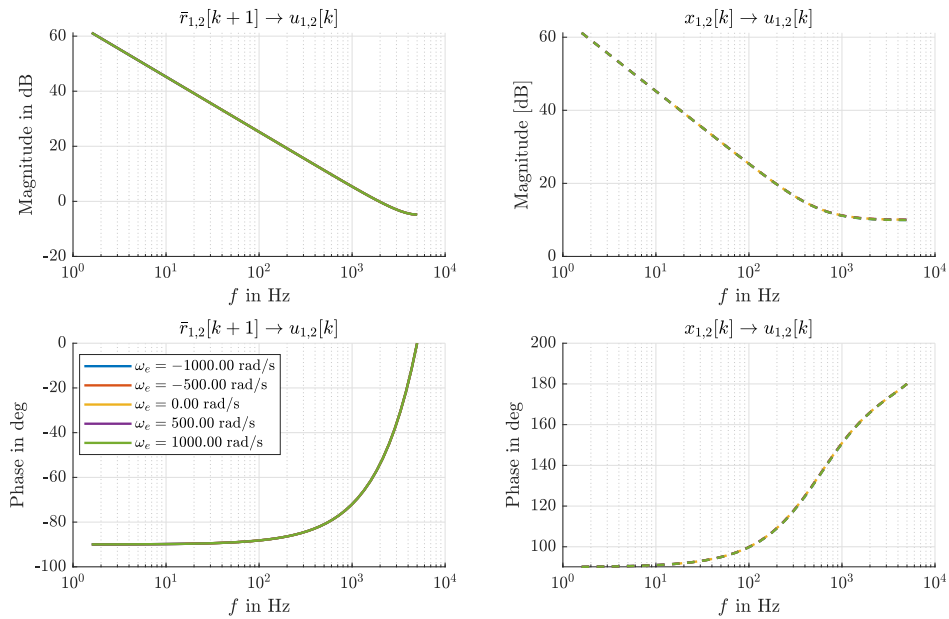


Figure 3.8: Frequency response for the unconstrained Hybrid MPC.

3.4.2 Closed-Loop Structure with active Constraints

3.4.3 Prediction Error Propagation

3.4.4 Integral Behavior

3.5 Extensions

3.5.1 Deadtime Compensation

3.5.2 Integral State Augmentation for the Basic MPC

Windup

3.5.3 Soft Constraints

3.6 Experimental Evaluation

3.6.1 Test Setup

3.6.2 Step Response

3.6.3 Constrained Operation

3.7 Comparison of the MPC Approaches

3.7.1 Steady State Behavior

3.7.2 Timing Uncertainties

3.7.3 Constrained Operation

3.7.4 Computational Effort

3.8 Discussion

Analyzing $\Sigma : \left(\mathbf{A}_c(\omega(t)), \tilde{\mathbf{B}}_c \right)$ in the context of linear parameter-varying systems, the framework of LPV System according to [10], one has to quantify the influence of the parameter ω . Due to the balance of angular momentum of the motor shaft (2.2), the electrical rotor speed $\omega = \omega_e = \omega_{m1}p$ is Lipschitz-continuous. With the eigenvalues of the respective dynamical system (2.2) being slower than (3.1), as discussed in Section 2.2.

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Appendix Title

This is an example of an appendix.